

Physics Mock Test

Current Electricity

Class 12 | CBSE / Board Prep

Marks: 24

Time: 60 Minutes

Instructions: This test covers Ohm's Law, Resistivity, Kirchhoff's Laws, Wheatstone Bridge, Potentiometer, and Heating Effects. Attempt all questions. For numericals, carry out calculations clearly.

Section 1: Multiple Choice Questions (1 Mark Each)

1. Ohm's Law

According to Ohm's law, $V = IR$. If the resistance of a conductor is doubled while the current remains constant, the potential difference across it:

- a) Doubles
- b) Halves
- c) Remains the same
- d) Becomes zero

2. Nature of Resistivity

The resistivity of a specific metallic wire depends primarily on:

- a) Length and Area
- b) Temperature and Material
- c) Current through it
- d) Applied Voltage

3. Series Resistance

In a series combination of three resistors R_v , R_r , and R_f , the equivalent resistance is always:

- a) Sum of individual values
- b) Product of individual values
- c) Less than the smallest
- d) Zero

4. Kirchhoff's Junction Rule

Kirchhoff's first law ($\Sigma I = 0$) at a junction is a physical statement of the law of conservation of:

- | | |
|-------------|-----------|
| a) Energy | b) Charge |
| c) Momentum | d) Mass |

5. Meter Bridge / Wheatstone Condition

In a balanced Wheatstone bridge consisting of arms P, Q, R, and S (in sequence), the balance condition is:

- | | |
|----------------------------|----------------|
| a) $P/Q = R/S$ | b) $P/Q = S/R$ |
| c) $P \cdot Q = R \cdot S$ | d) $P+Q = R+S$ |

6. Joule's Heating Effect

If the current passing through a fixed resistor is doubled, the rate of heat production (Power) becomes:

- | | |
|------------|------------------------|
| a) 2 times | b) 4 times |
| c) 8 times | d) $\frac{1}{2}$ times |

Section 2: Numerical & Short-Answer (2-3 Marks Each)

7. Basic Circuit Calculation

A 10 V battery with negligible internal resistance is connected to a resistor of $5\ \Omega$. Calculate the current flowing through the circuit.

8. Dimensions and Resistivity

A wire of length 2 m has a resistance of $10\ \Omega$. If its resistivity is $1.7 \times 10^{-8}\ \Omega\cdot\text{m}$, calculate the cross-sectional area of the wire.

9. Circuit Reconfiguration

Two resistors of $4\ \Omega$ and $6\ \Omega$ are connected to a 12 V battery. Find the total current drawn from the battery when they are connected in (i) Series and (ii) Parallel.

10. Parallel Branch Currents

A 6 V battery is connected to a parallel combination of a $2\ \Omega$ and $3\ \Omega$ resistor. Calculate the total current and the individual current passing through the $2\ \Omega$ resistor.

11. Unknown Resistance in Bridge

In a Wheatstone bridge, the values are $P = 10\ \Omega$, $Q = 15\ \Omega$, and $R = 20\ \Omega$. If the bridge is balanced, calculate the value of the unknown resistance S .

12. Practical Power Application

An electric bulb is rated 100 W, 220 V. Calculate:

- (i) The resistance of the filament.
- (ii) The current drawn when connected to a 220 V supply.
- (iii) The energy consumed in 1 hour.

Answer Key & Detailed Solutions

1. a) Doubles ($V = IR$, if R becomes $2R$, V becomes $2V$)

2. b) Temperature and nature of material (Independent of geometry)

3. a) Sum of individual resistances ($R_{eq} = R_1 + R_2 + R_3$)

4. b) Conservation of charge (Total current entering a junction = Total current leaving)

5. a) $P/Q = R/S$ (Standard null point condition)

6. b) 4 times ($H = I^2Rt$; if $I \rightarrow 2I$, then $I^2 \rightarrow 4I^2$)

7. $I = 2 \text{ A}$

$$I = V / R = 10 / 5 = 2 \text{ A.}$$

8. $A = 3.4 \times 10^{-9} \text{ m}^2$

$$\text{Using } R = \rho L / A \rightarrow A = \rho L / R = (1.7 \times 10^{-8} \times 2) / 10 = 3.4 \times 10^{-9} \text{ m}^2.$$

9. Series: 1.2 A | Parallel: 5 A

(i) Series: $R = 10\Omega$, $I = 12/10 = 1.2\text{A}$. (ii) Parallel: $R = (4 \times 6)/(4+6) = 2.4\Omega$, $I = 12/2.4 = 5\text{A}$.

10. $I_{\text{total}} = 5 \text{ A}$ | $I_{2\Omega} = 3 \text{ A}$

Combined $R = 1.2\Omega$. $I_{\text{total}} = 6/1.2 = 5\text{A}$. $I(2\Omega) = V/R = 6/2 = 3\text{A}$.

11. $S = 30 \Omega$

$$P/Q = R/S \rightarrow 10/15 = 20/S \rightarrow S = (20 \times 15) / 10 = 30 \Omega.$$

12. (i) 484Ω | (ii) 0.45 A | (iii) $3.6 \times 10^5 \text{ J}$

(i) $R = V^2/P = 220^2 / 100 = 484 \Omega$. (ii) $I = P/V = 100/220 \approx 0.45\text{A}$. (iii) $E = P \times t = 100\text{W} \times 3600\text{s} = 360,000 \text{ J}$.